IBM Z DATATHON 2026 · THEME 1 REAL-TIME AI FOR CRITICAL DECISIONS + THEME 2 AI SECURED

SepsisWatch Z

Early sepsis warning for the ICU, running entirely on IBM LinuxONE (z17): predictive AI on the Telum II NNPA, generative AI with IBM Granite, and CPACF hardware security. Patient data never leaves the mainframe.

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github.com/HemsyCA/DATAIBMZOS-IBM-Z-Datathon-2026 · Event code SSF2026

THE PROBLEM

Sepsis kills fast, and today's ICU alarms are too noisy to trust

~11M

sepsis-related deaths per year, about 1 in 5 deaths worldwide (Rudd et al., Lancet 2020)

every hour

of delayed treatment is associated with higher mortality (Kumar 2006; Seymour, NEJM 2017)

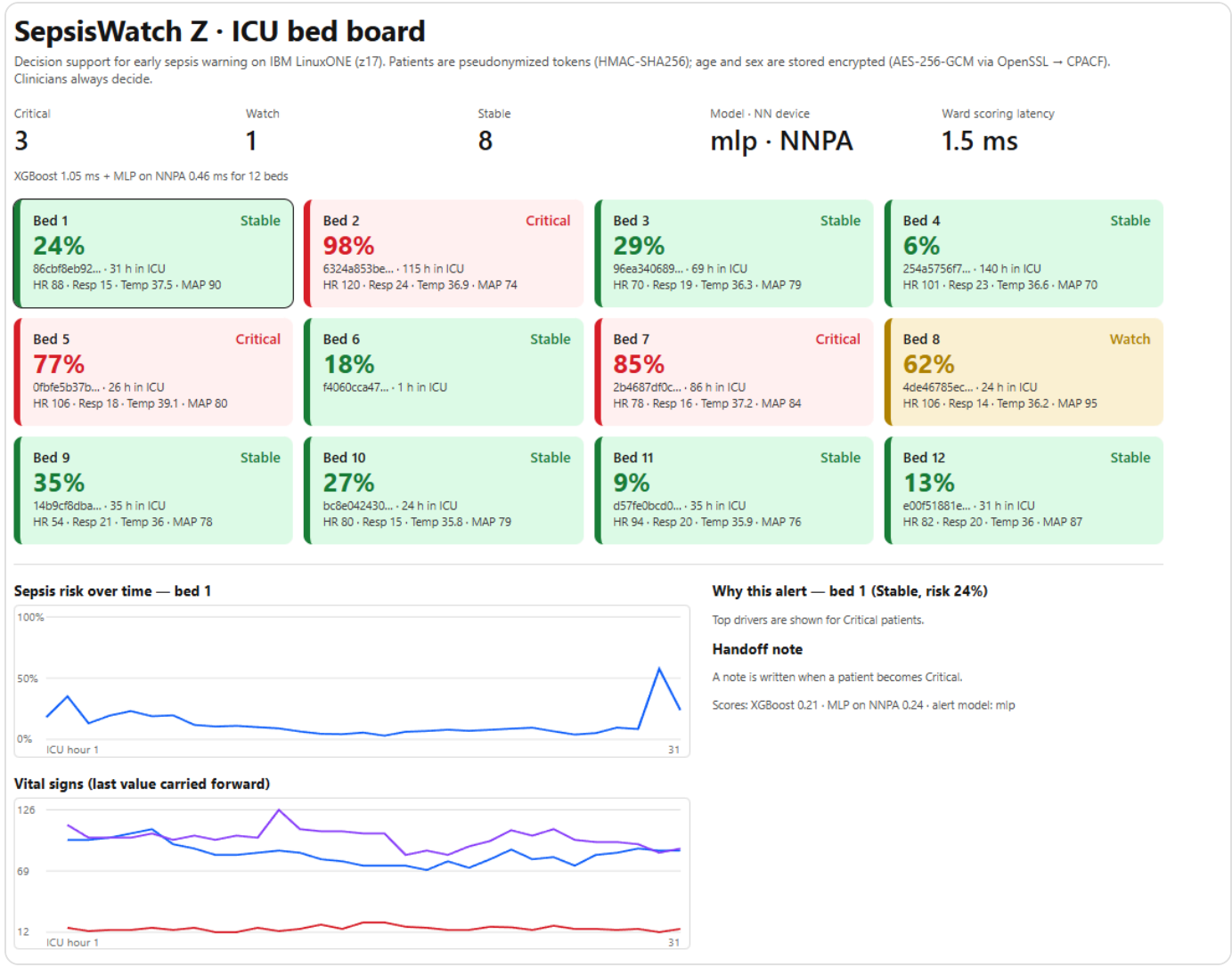
24%

of the time, SIRS ≥ 2 kept patients **without** sepsis in alarm on our test hospital → alarm fatigue

- Early signs are subtle and scattered across dozens of vitals and labs (lactate is missing in 97% of hours).
- ICUs with few specialists cannot watch every trend of every bed, every hour.
- Health data is highly sensitive: sending it to public-cloud AI is often not an option.

THE SOLUTION

One bed board: who to look at first, why, and a handoff note

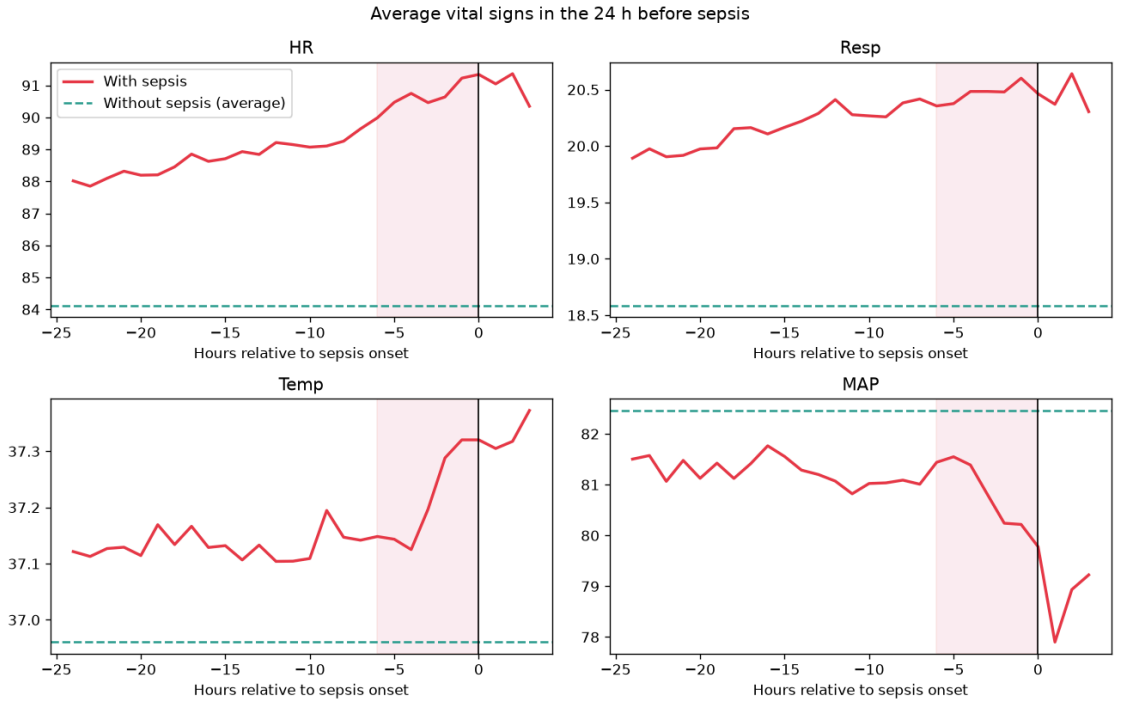


- **Every patient, every hour**: a risk score and a level: **Critical** / **Watch** / **Stable**.
- **Why this alert**: top clinical drivers (SHAP), physiology only.
- **Handoff note** by IBM Granite on the z17, **fact-checked** against the data before a nurse sees it.
- Whole ward scored in **~1.5 ms** (MLP on the NNPA + XGBoost).
- Decision support, not diagnosis: **clinicians always decide**.

DATA & METHOD

Real ICU data, validated on a hospital the model never saw

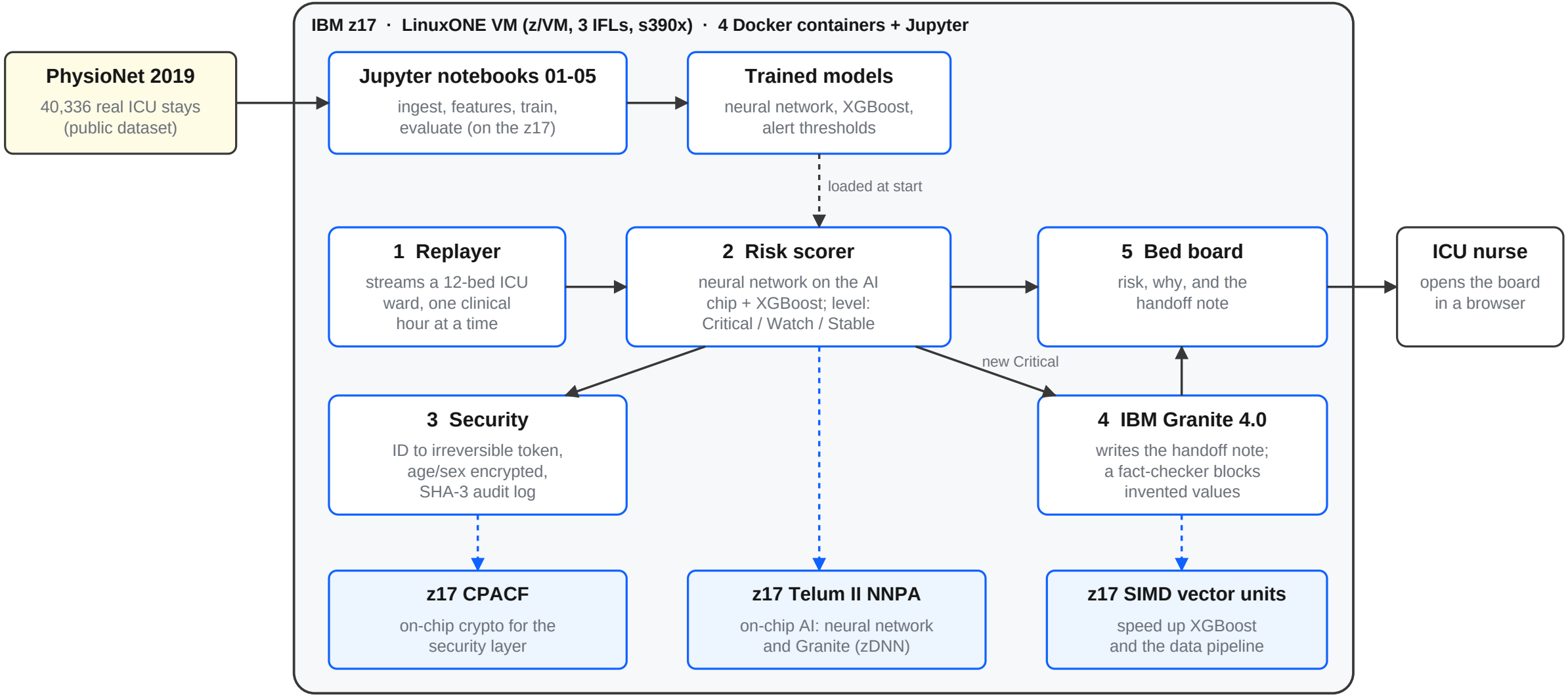
- **PhysioNet 2019 Sepsis Challenge**: 40,336 real ICU stays, 1,552,210 patient-hours, 40 variables.
- **Train on hospital A** (16,269 train / 4,067 validation patients) → **test on hospital B** (20,000 patients, 1,142 with sepsis).
- **140 hourly features** using only past data: last values, measurement patterns, 6-h trends, SIRS, qSOFA, shock index.
- **Models**: XGBoost + neural network (256-128-64) whose inference runs on the NNPA; SIRS/qSOFA baselines.
- **Alarm-aware thresholds**, chosen on validation: Critical ≤ 1 new false alarm per patient-day **and** ≤ 10% of non-sepsis hours.



Vital-sign trends before sepsis onset (notebook 01, run on the z17).

ARCHITECTURE

Everything runs inside one IBM z17: patient data never leaves



Detailed UML deployment and component diagram: appendix (slide 11) and docs/figures/architecture.png in the repository.

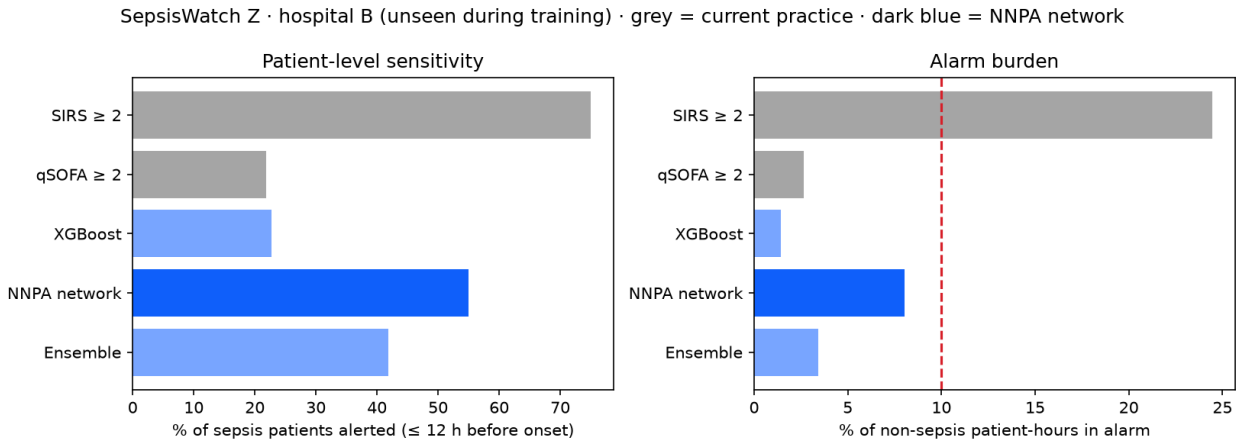
RESULTS · HOSPITAL B (UNSEEN)

Useful detection with one third of SIRS's alarm time

55%
of sepsis patients alerted up to 12 h before onset

8% vs 24%
non-sepsis patient-hours in alarm vs SIRS ≥ 2

−63%
new false alarms (0.49 vs 1.32 per patient-day)



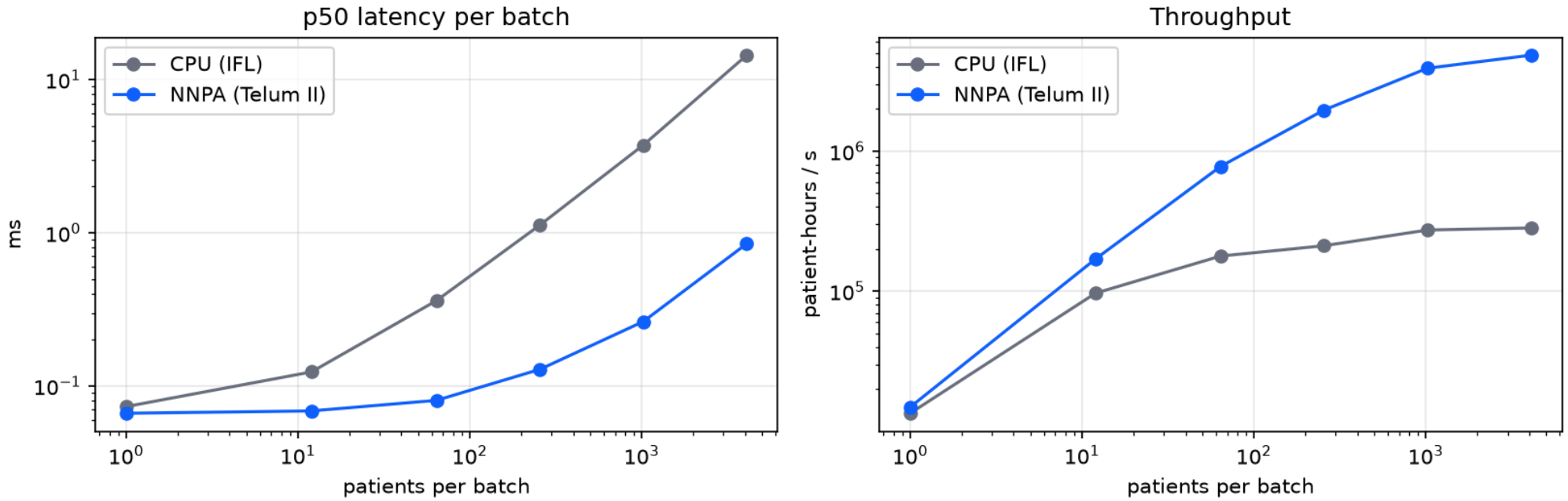
Model	Sens.	Alarm h	AUROC
SIRS ≥ 2	75.0%	24.4%	0.665
qSOFA ≥ 2	21.8%	2.7%	0.578
XGBoost	22.8%	1.4%	0.746
NNPA network	**55.0%**	**8.0%**	**0.711**
Ensemble	41.9%	3.4%	0.747

At near-SIRS sensitivity (68% vs 75%), the NNPA network still spends 40% less time in alarm (14.6% vs 24.4%). Model chosen on validation, not on the test set.

IBM Z PERFORMANCE

Telum II on-chip AI: faster at scale, identical decisions

SepsisWatch Z · MLP [256, 128, 64] (77,057 parameters) · hospital B · IBM z17



12×

whole hospital B (762k patient-hours): 0.22 s on NNPA vs 2.73 s on CPU

17×

4,096 patients per batch: 0.85 ms vs 14.4 ms · 12-bed ward in 0.07 ms

99.98%

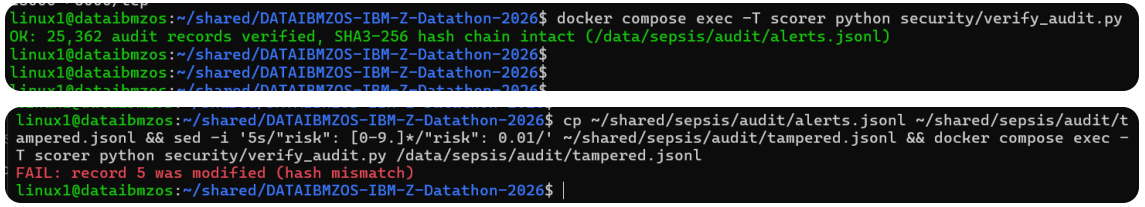
identical alert decisions CPU vs NNPA · same AUROC (0.711)

AI SECURED · TRUST BY DESIGN

Hardware security and a generative AI that cannot make things up

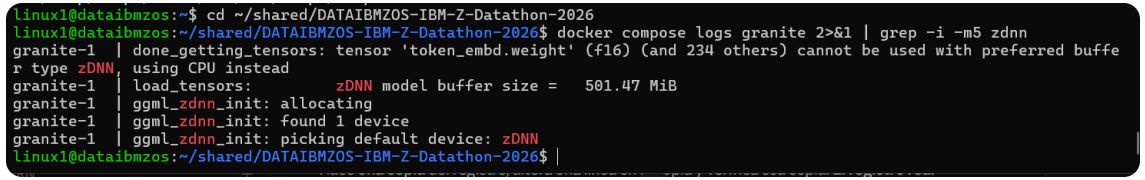
Security on IBM Z

- Patient IDs → **HMAC-SHA256 tokens**; key stays on the VM.
- Age/sex → **AES-256-GCM via OpenSSL → CPACF**.
- Every alert → **SHA3-256 hash chain**; edit one record and verification fails.
- No external API, no cloud LLM; ports bound to localhost.



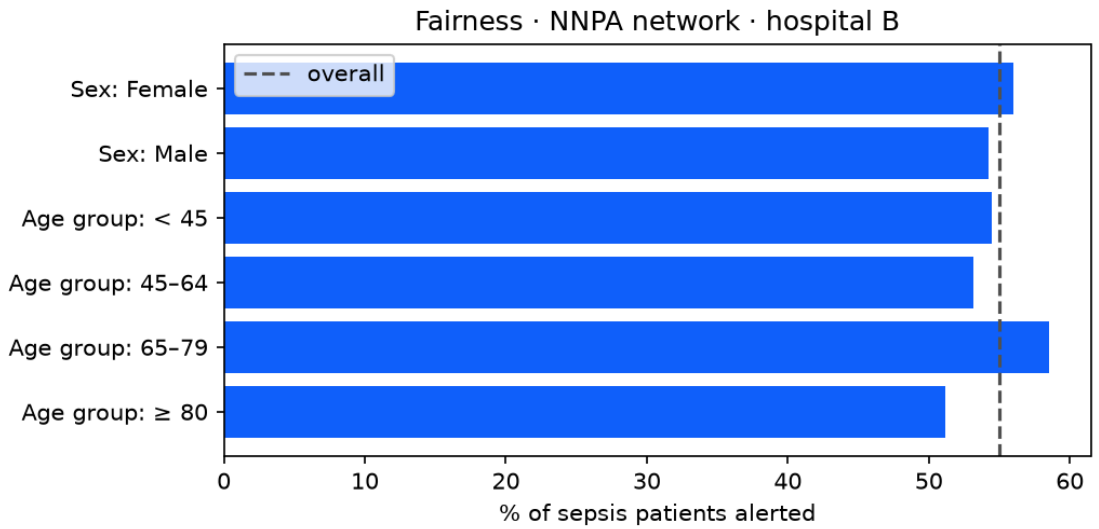
IBM Granite + fact-checker

- Granite 4.0 runs **on the z17** via llama.cpp's **zDNN** backend; about 7 s per note.
- Gets de-identified facts only; the bed number is set by the system.
- A note is rejected if it adds a number or topic not in the data, contradicts high/low, invents history or mentions drugs.
- Rejected → a deterministic template. **A nurse never sees an invented value.**



RESPONSIBLE AI

Fair across groups, honest about limits



Sensitivity by group, NNPA network, hospital B: gap 1.7 points by sex, 7.4 by age (lowest ≥ 80).

- Thresholds cap false alarms and alarm time to fight alarm fatigue.
- Explanations use physiology only; measurement-pattern features are never shown to nurses.
- A **risk score**, not a calibrated probability; decision support only.
- **Limits**: public 2019 US benchmark; replayed, not live; SIRS still catches more patients (75% vs 55%) at three times the alarm time; small Granite model.

IMPACT & NEXT STEPS

From a datathon prototype to a hospital pilot

Who benefits

- **ICU nurses & physicians**: know where to look first; alarms worth trusting.
- **Patients**: earlier attention to those deteriorating.
- **Hospitals**: on-premise AI with hardware encryption and an auditable trail; one IBM Z can serve many wards or hospitals.

Next steps

- Local validation and threshold tuning with partner hospitals.
- HL7 / FHIR integration with the EHR and bedside monitors.
- Larger or quantized Granite model for richer, faster notes.
- OpenShift on IBM Z for multi-hospital scale.

Built Oct 17–18, 2026 on IBM LinuxONE Community Cloud. Repo with 5 executed notebooks, 4 services and evidence screenshots: github.com/HemsyCA/DATAIBMZOS-IBM-Z-Datathon-2026

Telum II NNPA | zDNN | CPACF | z/VM | s390x containers | IBM Granite | PyTorch | XGBoost | FastAPI

APPENDIX · DETAILED ARCHITECTURE (UML DEPLOYMENT & COMPONENT DIAGRAM)

